

PRECISION AND RELIABILITY COSTS IN CONTRACTING CONTROL SYSTEMS

AUTHOR INFORMATION PENDING

ABSTRACT. We study finite catalogues of open-loop controls that keep trajectories within a shrinking neighbourhood of a constraint. The initial state selects a catalogue entry, and failure is measured by uniform area on the constraint. For a class of binary planar systems contracting toward the origin, we prove that the exponential catalogue rate at precision $\exp(-\gamma n + o(n))$ and failure probability $\exp(-\alpha n + o(n))$ is $\min\{S(\gamma), L_a(\alpha)\}$, provided the controlled maps have no critical points on the constraint. Here S is the full-state precision spectrum determined by the matched first derivatives, and L_a is the binary source-coding rate at a prescribed reliability exponent. The proof bounds probability transport through all feasible transient branches and constructs positive-area regions that force prescribed control prefixes under approximate feasibility. We also give an injective C^2 example with null critical sets that preserves every finite-horizon full-state catalogue count but has a strictly greater reliability cost. Thus preservation of full-state costs does not ensure preservation of exponential reliability. Under the weaker null-critical-set hypothesis, we also determine the full-state precision spectrum and the rate of one fixed, nearly full-area compact initial set.

1. INTRODUCTION

At time zero, a controller observes the initial state and selects one open-loop input from a finite catalogue. How large must that catalogue be to maintain a state constraint? Its size measures the information needed to make this initial choice. Invariance entropy describes its exponential growth with the horizon; see [5, 8]. When all inputs contract toward a common state, a fixed positive tolerance eventually makes the constraint automatic, and the required catalogue size stays bounded. Shrinking the tolerance with the horizon reveals a nonzero cost. Allowing failure on a set of small initial-state probability introduces a second scale, whose interaction with precision is the subject of this paper.

We use the triangle $Q = \{(s, z) : 0 \leq s \leq 1, |z| \leq s\}$ with uniform area probability. An input succeeds at an initial state if its trajectory stays within Euclidean distance δ_n of Q at every time $0, \dots, n$. It may leave and reenter Q while satisfying this condition. Every binary input is available, and its assignment depends on the actual initial state. We study precision and failure sequences satisfying

$$\delta_n \rightarrow 0, \quad \delta_n = \exp(-\gamma n + o(n)), \quad e_n = \exp(-\alpha n + o(n)), \quad \gamma \geq 0, \quad \alpha > 0.$$

The successful set may vary with n . We also compare this requirement with covering a compact initial set fixed before all horizons and precision sequences.

The systems are globally C^2 , are controlled invariant on Q , and contract toward the origin in a common norm. Their first derivatives induce complementary angular branches with relative lengths a_0, a_1 , where $a_0 + a_1 = 1$, and matched radial factors a_i^β , $\beta > 1$. Radial motion may depend on angle; the nonlinear feasible branches may overlap and have truncated images. Section 2 states the full assumptions, including the noncollapse conditions used for the different conclusions.

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1.1. Main results and proof mechanism. If neither controlled map has a critical point on Q , Theorem 2.1 gives

$$\lim_{n \rightarrow \infty} n^{-1} \log r_{e_n}(n, \delta_n; Q) = \min\{S(\gamma), L_a(\alpha)\}.$$

The two terms come from different constructions. Covering all initial states at the prescribed tolerance gives the precision spectrum S . Keeping exactly feasible words outside a sufficiently rare family gives L_a , the classical binary source-coding rate at reliability exponent α . The smaller catalogue suffices. The lower bound shows that allowing arbitrary approximate inputs cannot improve its exponential rate. Both functions are evaluated explicitly. For $\gamma \leq \beta h$, where $h = -\sum_i a_i \log a_i$, the rate is γ/β regardless of $\alpha > 0$. Above this threshold, exponentially reliable coverage costs strictly more than coverage with a fixed positive failure allowance, until the full-state rate is reached. The transmitted index selects a whole open-loop input at time zero; no subsequent communication is considered.

Ordinary precision costs require a weaker hypothesis. If the critical sets merely have area zero, the full-state rate remains S and every positive-area compact initial set has rate γ/β below βh . There is a single compact set of area arbitrarily close to one with rate $\min\{h, \gamma/\beta\}$ for all finite precision exponents (Theorem 2.3). Theorem 2.2 shows why this weaker noncollapse does not suffice for reliability: an injective C^2 transient preserves every finite-horizon full-state catalogue count of the linear model while increasing its reliability cost. The transient changes the probability of states requiring many different inputs.

The main difficulty is to pass from control words to the states that an arbitrary approximate input actually covers. At a fixed radius, a complete feasible section may be empty and its terminal image may be truncated. Several controls may also be feasible at the same state, so disagreement with a designated coding need not violate the constraint. We instead partition covered states by their first actual exit from Q . An angular product measures the width of the preceding feasible region, while a radial product measures how much exit the tolerance permits. Their ratio yields Theorem 2.5. After a finite transient, this estimate holds on one fixed compact set with arbitrarily small area loss.

An exponentially small failure allowance cannot absorb an arbitrary fixed area loss, or even an error of order δ_n . Two further estimates resolve this issue. Lemma 7.1 bounds probability transport through all feasible transient branches by one constant independent of the horizon and both budgets. Lemma 7.2 supplies disjoint positive-area regions of known mass on which any successful input must follow a prescribed prefix. At the first wrong control, the distance from Q exceeds the tolerance; later reentry cannot undo that violation. These estimates supply the physical mass bounds needed to apply type counting to the optimal catalogue.

1.2. Related work. Our full-cover quantity is the outer invariance complexity of Chen and Zhong [2, Section 2.2], with a one-step change of horizon convention. Their Section 3 also considers measure-dependent catalogue complexity. These frameworks permit noninjective continuous controlled maps. Their boundedness results at fixed tolerances concern the fixed-scale problem; the proofs do not estimate how the resulting constants depend on simultaneous exponentially decreasing precision and failure allowances.

Wang, Huang and Sun [12] define measure invariance entropy through invariant partitions and cylinder probabilities. Their Theorem 4.8 transports the initial measure together with a control conjugacy. Colonius [3] relates quasi-stationary metric invariance entropy to coder-controllers, and Wang and Huang [11] study spanning-based measure entropy and inverse variational principles. Probability-dependent control catalogues and transport of initial measures are therefore established parts of the subject.

Nie, Wang and Huang [10] obtain measure entropy through convex duality from invariance pressure. Chen, Huang and Zhong [1] study entropy dimension using invariant-partition cylinders and time-power normalisation, while Yang, Chen, Yang and Zhou [13]

develop cost-threshold pressure formulas. Applying these methods to the present problem requires a quantitative comparison between symbolic weights and the physical sets covered by arbitrary approximate inputs. In particular, transporting a measure does not compare two systems both endowed with uniform area. Our probability domination and forced-region estimates provide this comparison for the stated structural class. The transient counterexample shows that null-set preservation alone cannot supply it.

Volume estimates for the set served by a single control already give catalogue lower bounds in the invariance-pressure setting; see Colonius, Cossich and Santana [4]. We use this general principle together with classical graph estimates, summable distortion, stopping scales and typical sets; related geometric methods appear in [7, 9]. The entropy optimisation is an application of the method of types [6]. Here the estimates must retain the tolerance at the first true exit and the mass of regions whose failure cannot be absorbed by e_n . The area and probability comparisons above establish that connection. Once they are available, the explicit rates follow by counting.

1.3. Organisation. Section 2 defines the catalogues and states the results. Sections 3–5 establish feasible-region bounds, transient area control and trajectories with a wrong-control margin. Section 6 derives the ordinary precision spectrum, and Section 7 proves the reliability formula. Sections 8 and 9 give the transient counterexample and an example with inward boundary images. Appendix A characterises first-order matching within the specified matrix family. Appendix B distinguishes finite-horizon counts from rates when the control alphabet is enlarged.

2. CONTROL CATALOGUES AND MAIN RESULTS

2.1. Catalogue and precision. Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For an infinite binary input $u = (u_1, u_2, \dots)$ set $F_u^0 = \text{id}$ and $F_u^k = F_{u_k} \circ \dots \circ F_{u_1}$. All binary inputs are allowed. Distances are Euclidean, and all logarithms are natural. For a nonempty compact $K \subset Q$ define

$$(2.1) \quad A_u(n, \delta; K) = \{x \in K : \text{dist}(F_u^k x, Q) < \delta \text{ for every } 0 \leq k \leq n\}.$$

Let $r_F(n, \delta; K)$ be the smallest number of infinite inputs whose sets (2.1) cover K . The assignment may depend on the actual initial state. Exit and reentry are allowed subject to every displayed constraint. Only the first n letters affect this count. Controlled feasibility makes it finite: all binary words of length n , with arbitrary infinite continuations, already provide a catalogue.

A precision sequence means $\delta_n > 0$, $\delta_n \rightarrow 0$ and $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone. With the convention $[0, N)$ used in [2, Section 2.2], the exact correspondence is $r_F(n, \delta; K) = r_{\text{inv}}(n+1, \delta, K, Q)$.

Let $\mu = \text{area}|_Q$, which is a probability since $\text{area}(Q) = 1$. For $0 < e < 1$, define

$$(2.2) \quad r_e(n, \delta; Q) = \min \left\{ \#\mathcal{U} : \mu \left(\bigcup_{u \in \mathcal{U}} A_u(n, \delta; Q) \right) \geq 1 - e \right\}.$$

The minimum is over finite catalogues of unrestricted binary inputs. It exists because the full-cover count is finite. A failure sequence has $0 < e_n < 1$, $e_n \rightarrow 0$ and $-\log e_n/n \rightarrow \alpha > 0$. Both sequences may be nonmonotone. The successful physical subset in (2.2) may change with n ; it is not a fixed compact initial set.

2.2. First-order data and global assumptions. Fix

$$(2.3) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta_0, \beta_1 > 1,$$

and set $r_i = a_i^{\beta_i}$, $t_i = r_i/a_i < 1$, $a_* = \min_i a_i$, $a^* = \max_i a_i$ and $h = -\sum_i a_i \log a_i$. Write

$$(2.4) \quad A_0 = \begin{pmatrix} r_0 & 0 \\ t_0 a_1 & t_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} r_1 & 0 \\ -t_1 a_0 & t_1 \end{pmatrix}.$$

We retain separate radial exponents for the geometric estimates and the comparison in Appendix A. The reliability theorem uses their matched subfamily. The standing assumptions on the actual maps are:

- (i) $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are C^2 maps, $F_i(0) = 0$ and $DF_i(0) = A_i$.
- (ii) $Q \subset F_0^{-1}Q \cup F_1^{-1}Q$.
- (iii) For some $\Lambda \geq 1$ and $0 < \theta < 1$, the norm $V(s, z) = \max\{\Lambda|s|, |z|\}$ satisfies

$$(2.5) \quad V(F_i x) \leq \theta V(x) \quad (x \in \mathbb{R}^2, i = 0, 1).$$

- (iv) Each critical set

$$(2.6) \quad \mathcal{C}_i = \{x \in Q : \det DF_i(x) = 0\}$$

has area zero.

The second assumption is one-step controlled invariance; unrestricted concatenation yields an exactly feasible infinite input from each point of Q . The third concerns contraction toward the origin, not distances between two arbitrary trajectories. In particular it controls appended tails even outside Q . The fourth supplies noncollapse along feasible finite transients; it allows folds and requires no condition on critical points outside Q . The maps may therefore be noninjective on Q , and radial motion may depend on angle. The coverage estimates below are consequences of these four assumptions.

For comparison, a fixed tolerance has zero exponential cost. Indeed $\|x\|_2 \leq \sqrt{2}V(x)$, and (2.5) gives a fixed J_δ such that every input sends Q into the δ -ball about zero after J_δ steps and keeps it there. Choose an exactly feasible prefix of that length for each initial state and then append an arbitrary tail. The at most 2^{J_δ} prefixes cover Q for every longer horizon. Shrinking precision is what prevents this finite-transient argument from making the asymptotic question trivial.

Both matched and mismatched subfamilies are nonempty. For the linear maps A_i , the angular domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover every fibre. Put $t = \max_i t_i < 1$ and choose $\Lambda = 1 + 4t/(1 - t)$. Then $V(A_i x) \leq \max\{\max_i r_i, t(1 + 1/\Lambda)\}V(x)$ globally, with factor less than one, and $\det A_i = r_i^2/a_i > 0$.

Two products will be used throughout the geometric estimates:

$$\lambda_w = \prod_{j=1}^{|w|} a_{w_j}, \quad P_w = \prod_{j=1}^{|w|} r_{w_j},$$

with empty products one. The angular product λ_w controls the width of a feasible initial section; the radial product P_w controls the physical scale reached after the word. Their comparison is proved in Section 3, without assuming that a terminal section fills the angular interval. We also use the mean radial cost

$$(2.7) \quad \chi_R = -\sum_i a_i \log r_i > h.$$

2.3. Probability costs. Write H_b for binary entropy. In the matched case $\beta_0 = \beta_1 = \beta$, the angular cost and spectrum are

$$(2.8) \quad H_b(p) = -p \log p - (1 - p) \log(1 - p), \quad \chi(p) = -p \log a_0 - (1 - p) \log a_1,$$

$$(2.9) \quad S(\gamma) = \max_{p \in [0,1]} H_b(p) \min\{1, \gamma/[\beta \chi(p)]\},$$

where $0 \log 0 = 0$. In particular $H_b(a_0) = \chi(a_0) = h$.

$$(2.10) \quad I_a(p) = \chi(p) - H_b(p), \quad L_a(\alpha) = \max_{I_a(p) \leq \alpha} H_b(p).$$

The function I_a is relative entropy with respect to (a_0, a_1) . For reference words with zero frequency p , $H_b(p)$ gives the count exponent and $I_a(p)$ the probability exponent. Thus L_a is the classical binary source-coding rate at reliability exponent α . The theorem below identifies how this word count enters the physical control problem.

Theorem 2.1 (Precision–reliability transfer). *Assume the standing hypotheses with $\beta_0 = \beta_1 = \beta$, and strengthen the critical-set condition to*

$$(2.11) \quad \det DF_i(x) \neq 0 \quad (x \in Q, i = 0, 1).$$

For every fixed such system, every finite $\gamma \geq 0$, every $\alpha > 0$, and every precision and failure sequence of those exponents,

$$(2.12) \quad \lim_{n \rightarrow \infty} \frac{\log r_{e_n}(n, \delta_n; Q)}{n} = \min\{S(\gamma), L_a(\alpha)\}.$$

No global injectivity or global inverse is required.

For a finite catalogue, enumerate its inputs and assign a successful state to its first successful input. The sets are Borel, so this is a measurable initial-state encoder. A fixed-length index sent once at time zero has minimum bit length $\lceil \log_2 r_e \rceil$; the asymptotic bit rate is (2.12) divided by $\log 2$. There are no later messages: this is the cost of one initial index. It gives neither a causal feedback data-rate nor a channel capacity.

Theorem 2.2 (A null-critical transient changes reliability). *For $a_0 = 2^{-10}$, $a_1 = 1 - 2^{-10}$ and $\beta = 4096$, there is a fixed injective C^2 pair satisfying the standing matched hypotheses, with critical sets of area zero, such that*

$$(2.13) \quad r_F(n, \delta; Q) = r_A(n, \delta; Q) \quad (n \geq 1, \delta > 0).$$

It maps no positive-area subset of Q into an area-zero set. Nevertheless, for $\gamma = -\beta \log(a_0 a_1)$ and $\alpha = 1/8$, every pair of sequences with these exponents satisfies

$$(2.14) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \frac{1}{2} \log 2 > \min\{S(\gamma), L_a(\alpha)\}.$$

Condition (2.11) is a sufficient condition for (2.12). The counterexample rules out replacing it merely by critical sets of area zero in a universal transfer theorem. It does not show that every system with critical points fails, or give a necessary and sufficient probability criterion.

2.4. Ordinary precision costs and fixed initial sets. For full coverage, and for coverage of a fixed compact initial set, the standing null-critical hypothesis suffices. The next theorem provides the spectrum used in Theorem 2.1 and shows why the choice of a large initial set is a different question from an exponentially small failure allowance.

Theorem 2.3 (Precision spectrum and initial-set dependence). *Suppose $\beta_0 = \beta_1 = \beta$ in the standing assumptions. For every fixed system, every finite $\gamma \geq 0$ and every precision sequence of exponent γ ,*

$$(2.15) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = S(\gamma).$$

Every fixed positive-area compact $K \subset Q$ satisfies

$$(2.16) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq S(\gamma).$$

Thus the rate exists and is γ/β for $0 \leq \gamma \leq \beta h$. For each $\zeta \in (0, 1)$ there is one fixed compact $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$ such that, simultaneously for every finite $\gamma \geq 0$ and every precision sequence of that exponent,

$$(2.17) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \min\{h, \gamma/\beta\}.$$

The set is chosen before the exponent, sequence and horizon.

Corollary 2.4 (Closed spectrum and two thresholds). *Under the hypotheses of Theorem 2.3, put $a = a_* \in (0, 1/2)$, $c_b = -\log(1 - a)$ and $d_b = \log((1 - a)/a) > 0$. The common-cost boundary and saturation boundary are respectively*

$$(2.18) \quad \gamma_* = \beta h, \quad \gamma_{\text{sat}} = -\frac{\beta}{2} \log(a_0 a_1) = \beta(c_b + d_b/2) > \gamma_*.$$

The full-set spectrum is

$$(2.19) \quad S(\gamma) = \begin{cases} \gamma/\beta, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b), & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

The exact gap between Q and the single set K_ζ in the theorem is

$$(2.20) \quad \Delta(\gamma) = S(\gamma) - \min\{h, \gamma/\beta\} = \begin{cases} 0, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b) - h, & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2 - h, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

It is strictly positive exactly when $\gamma > \gamma_*$. The full-set rate equals $\log 2$ exactly when $\gamma \geq \gamma_{\text{sat}}$.

The first threshold concerns equality across all positive-area compact initial sets; the second concerns the full-set catalogue. The rate of an arbitrary compact initial set above the first threshold is constrained by (2.16), but is not determined by the theorem.

The role of matching can also be tested within the same matrix family. Appendix A proves that unequal radial powers give distinct full-state and near-full-set rates throughout an interval of positive exponents adjacent to zero. A common positive-area rate on such an interval therefore holds precisely when the powers match. This criterion concerns ordinary precision costs in the specified family; it is not a classification of reliability.

2.5. The physical coverage estimate. The following estimate bounds the mass covered by one arbitrary input, without selecting an exact coding. It supplies the lower rate for positive-area initial sets in both cases. Its regular compact set also provides the single near-full set used in the upper bounds. These are fixed-loss estimates: their $C_T \delta$ term is not assigned to an exponentially small probability budget.

Theorem 2.5 (Area covered by an arbitrary approximate input). *Under the standing assumptions, whether matched or mismatched, there is a fixed integer $J \geq 1$ with the following property. For every positive-area compact $K \subset Q$ and every $0 < \varepsilon < \text{area}(K)$, one can choose a compact $T \subset K \setminus \{0\}$ with $\text{area}(T) > \text{area}(K) - \varepsilon$ and $C_T < \infty$ so that for every $\eta \in (0, h)$ there is $C_{T,\eta} < \infty$ such that*

$$(2.21) \quad \text{area}(T \cap A_u(n, \delta; K)) \leq C_T \delta + C_{T,\eta} \left[e^{-(h-\eta)m} + \delta e^{(\chi_R - h + 2\eta)m} \right]$$

for every input u , integers $n \geq J$ and $0 \leq m \leq n - J$, and $0 < \delta \leq 1$. The integer depends only on the fixed system. The set and C_T precede η, n, δ, u ; the second constant may also depend on η . In particular, for $K = Q$ the set can have area greater than $1 - \zeta$ for any prescribed $\zeta \in (0, 1)$.

3. LOCAL FEASIBLE GEOMETRY

This section proves two bounds used later: a complete feasible word region has area at most a constant times λ_w , and states surviving a first exit after w have area at most a constant times $\delta\lambda_w/P_w$. We obtain both from monotonicity along radial graphs. The first bound concerns exactly feasible prefixes; the second is what allows arbitrary approximate inputs in the catalogue lower bounds.

Use (s, y) with $z = sy$ for $s > 0$. All local estimates below are on $0 < s \leq \rho$, $|y| \leq 1$; their extensions at $s = 0$ are specified next. Define

$$R_i(s, y) = F_i^s(s, sy)/s, \quad N_i(s, y) = F_i^z(s, sy)/s, \quad T_i = N_i/R_i.$$

The integral formula $F_i(s, sy)/s = \int_0^1 DF_i(ts, tsy)(1, y) dt$ extends R_i, N_i as C^1 functions to $s = 0$. At zero

$$(3.1) \quad R_i(0, y) = r_i, \quad L_0(y) := T_0(0, y) = (y + a_1)/a_0, \quad L_1(y) := T_1(0, y) = (y - a_0)/a_1.$$

The local errors below are of order s . Radial contraction makes their sum finite along a feasible orbit, which will give product comparisons independent of the word length. We choose all constants from the maps before considering individual words. Let L bound the Euclidean bilinear operator norm of both second derivatives on $\bar{B}(0, 2)$, and let $r_* = \min r_i$, $r^* = \max r_i$, $a^* = \max a_i$, $a_* = \min a_i$. Take $B = 10^4(1 + L)/r_*^3$, and shrink $\rho > 0$ so that

$$(3.2) \quad \rho \leq \min \left\{ \frac{1}{4}, \frac{r_*}{8(1 + L)}, \frac{1 - r^*}{8B}, \frac{1/a^* - 1}{16B}, \frac{1}{16B}, \frac{a_*}{16B} \right\}.$$

Taylor's integral formula gives $|R_i - r_i| \leq 2Ls$, $|\partial_s R_i|, |\partial_s N_i| \leq 2L$, and $|\partial_y R_i| \leq 2Ls$, $|\partial_y N_i - t_i| \leq 2Ls$. The same bounds apply with the appropriate constant linear term in N_i . In particular $R_i \geq r_*/2$. Dividing these formulas by R_i , and differentiating the quotient, gives, with the common constant B ,

$$(3.3) \quad \begin{aligned} |R_i - r_i| &\leq Bs, & |\log(R_i/r_i)| &\leq Bs, \\ |\partial_s \log R_i| &\leq B, & |\partial_y \log R_i| &\leq Bs, \\ |T_i - L_i| &\leq Bs, & |\partial_s T_i| &\leq B, & |\partial_y T_i - 1/a_i| &\leq Bs. \end{aligned}$$

For example, $|N_i| \leq 3$ on the stated rectangle, and quotient derivatives use $R_i^{-2} \leq 4/r_*^2$. Thus the chosen B bounds every term in (3.3).

Set $\bar{r} = r^* + B\rho < 1$ and $\hat{r} = r^* + 2B\rho < 1$. On an exactly feasible local orbit, $s_k \leq \bar{r}^k s_0$. The bounds in (3.2) also give $T_1(s, -1) < -1$ and $T_0(s, 1) > 1$. Controlled invariance therefore forces

$$(3.4) \quad T_0(s, -1) \geq -1, \quad T_1(s, 1) \leq 1.$$

Equality is not required in (3.4). The increasing functions $T_i(s, \cdot)$ have unique cut points $\vartheta_0(s)$ with $T_0 = 1$ and $\vartheta_1(s)$ with $T_1 = -1$. They equal $a_0 - a_1 + O(s)$. Their feasible intervals are respectively $[-1, \vartheta_0(s)]$ and $[\vartheta_1(s), 1]$. Controlled invariance gives $\vartheta_1 \leq \vartheta_0$. Thus any local overlap has angular width $O(s)$ and physical width $O(s^2)$. Positivity of the overlap is not assumed.

3.1. Complete intervals with truncated images. Consider a radial graph $s = S(y)$, on any closed subinterval of $[-1, 1]$, with $|S'/S| \leq 1$. Put $k = S'/S$. For either input its terminal angular derivative along the graph is

$$J_i = \partial_y T_i + \partial_s T_i S' = 1/a_i + O(BS) > 0.$$

The new radial graph has logarithmic slope

$$(3.5) \quad k_{\text{new}} = \frac{k + (\partial_s \log R_i)S' + \partial_y \log R_i}{J_i}.$$

Its numerator is at most $1 + 2B\rho$ in absolute value and its denominator at least $1/a^* - 2B\rho$. By (3.2), $|k_{\text{new}}| \leq 1$. Piecewise C^1 graphs obey the same assertion almost everywhere.

For a fixed initial radius $0 < s_0 \leq \rho$, start with the horizontal graph $S \equiv s_0$. Let $I_w(s_0)$ be the set of initial angles whose trajectory under a finite word w stays in Q at every intermediate time. It is a closed interval, possibly empty or a point. To check every intermediate constraint, suppose the current angular image is an interval with a graph obeying (3.5). Extend that graph constantly to the two sides of its domain. The extension is positive, bounded by ρ , and obeys the logarithmic-slope bound almost everywhere. Both $T_i(S(y), y)$ are increasing. By (3.4), control 0 cannot violate the lower angular boundary, and control 1 cannot violate the upper one. The other boundary imposes one cut. Pulling that cut back through the increasing preceding angular map leaves an interval. No terminal map is asserted to be onto $[-1, 1]$. The two allowed children cover the parent, by controlled invariance, but either child can be empty.

Along any such feasible graph, (3.3) and the geometric decay of s_j give constants independent of w, s_0 and its initial angle:

$$(3.6) \quad C_R^{-1} s_0 P_w \leq s_{|w|} \leq C_R s_0 P_w, \quad C_A^{-1} \lambda_w^{-1} \leq \frac{dY_w}{dy_0} \leq C_A \lambda_w^{-1}.$$

For instance one may take $C_R = \exp(B\rho/(1 - \bar{r}))$, $C_A = \exp(4B\rho/(1 - \bar{r}))$: $|a_i J_i - 1| \leq 2Bs_j \leq 1/8$, so its logarithm has modulus at most $4Bs_j$. The first estimate sums $\log(R_i/r_i)$; the second sums the angular derivatives. They are independent product estimates. Only in the matched case is $P_w = \lambda_w^\beta$. Integration over the *actual* angular image implies

$$(3.7) \quad |I_w(s_0)| \leq 2C_A \lambda_w, \quad |\mathcal{I}_w| \leq C_A \rho^2 \lambda_w, \quad \mathcal{I}_w = \{(s, sy) : 0 < s \leq \rho, y \in I_w(s)\}.$$

The Jacobian of $(s, y) \mapsto (s, sy)$ is s . There is no positive lower bound on $|I_w|$; some fixed-radius slices in Section 9.1 are empty.

For a point interval the length bound is zero. Derivative comparisons on nondegenerate intervals extend to their endpoints by continuity; degenerate fibres remain in all covers and do not contribute area.

3.2. The first true exit. For a fixed input, partition its covered states by the first time at which its trajectory actually leaves Q . Up to that time the prefix is exactly feasible, so the preceding interval estimates apply. The permitted tolerance restricts the states that can survive this first exit.

Suppose the prefix w of length $j - 1$ is exactly feasible and the next prescribed input i is the first one to leave Q . On the current graph the map T_i is increasing with derivative at least $1/(2a_i)$. The radial output is at least $cs_0 P_w$, by (3.3), (3.6), with $c > 0$ fixed. If this wrong step nonetheless has distance less than δ from Q , then

$$(3.8) \quad (|z_j| - s_j)_+ < 2\delta.$$

This follows from the Euclidean Lipschitz linear functionals $z - s$ and $-z - s$. The output radius is positive and at most ρ , so the violated constraint is angular, not the upper radius. Fix s_0 and regard the current state as a point on the image graph after w . On the side where input i violates the constraint, (3.8) and the radial lower bound give

$$0 < |T_i(S(y), y)| - 1 < \frac{2\delta}{cs_0 P_w}.$$

Only one angular boundary can be crossed, by (3.4). The derivative lower bound for T_i along this graph therefore limits this set of current angles to length $C\delta/(s_0 P_w)$. Pulling it back by the preceding angular map in (3.6) gives initial-angle length at most

$$C\delta\lambda_w/(s_0 P_w).$$

Integration against $s_0 ds_0$ then bounds its physical area by

$$(3.9) \quad C\delta\lambda_w/P_w.$$

This argument also covers a point interval or a terminal interval lying wholly beyond the cut: monotonicity bounds the diameter of the part satisfying (3.8), without requiring a cut point inside the actual image. It does not require the wrong trajectory to remain outside afterwards. Any later reentry cannot erase constraint (3.8) at its first exit. A different exact feasible coding is not counted as an exit and has no role in (3.9).

4. FINITE TRANSIENTS AND ACTUAL COVERED AREA

Common contraction gives a uniform time of entry into the local region. To transport its area estimates back to the initial set, we remove a small area once and retain all feasible transient choices on the remaining compact window. This is where the null-critical-set hypothesis is used; a global inverse is not required.

For a finite word w , let

$$(4.1) \quad D_w = \{x \in Q : F_{w|k}x \in Q \ (1 \leq k \leq |w|)\}, \quad D_\emptyset = Q.$$

These sets are compact. Write w^- for the word with its last letter removed. At a first true exit under w , all derivative inputs still lie in Q , so the relevant domain is D_{w^-} . The last output need not belong to Q .

Lemma 4.1 (Null pullbacks along feasible prefixes). *For every Lebesgue null set $B \subset \mathbb{R}^2$, every control i , and every finite word w ,*

$$(4.2) \quad \text{area}(Q \cap F_i^{-1}B) = 0, \quad \text{area}(D_w \cap F_w^{-1}B) = 0.$$

For each nonempty word the compact set

$$(4.3) \quad Z_w = D_{w^-} \cap \{x : \det DF_w(x) = 0\}$$

also has area zero.

Proof. The inverse function theorem supplies an open inverse chart at every point of $Q \setminus \mathcal{C}_i$. A countable family of these charts covers that set. Exhaust each image by relatively compact subcharts; its C^1 inverse is Lipschitz on sufficiently small compact neighbourhoods. A Lipschitz map preserves null sets: cover a bounded null set by squares with arbitrarily small total area, and cover their images by discs whose total area is bounded by a constant times that sum. Thus every chart pullback of B is null. The omitted $\mathcal{C}_i$ is null by (2.6). A Borel null superset handles an arbitrary Lebesgue null set.

For $w = vi$, exact feasibility gives

$$D_w \cap F_w^{-1}B \subset D_v \cap F_v^{-1}(Q \cap F_i^{-1}B).$$

Induction proves the second assertion of (4.2), including the empty word. Finally, on D_{w^-} the chain rule expresses $\det DF_w$ as a product of determinants evaluated at states in Q . If the factor at time j vanishes, the initial state belongs to

$$D_{w|j-1} \cap F_{w|j-1}^{-1}\mathcal{C}_{w_j}.$$

This is null by (4.2). Their finite union contains Z_w , which is compact by continuity and closedness of the feasibility domain. $\square$

Choose $J \geq 1$, depending only on the fixed system, so that

$$(4.4) \quad \theta^J \sup_Q V \leq \rho.$$

Every exactly feasible J -prefix ends in $Q_\rho = Q \cap \{s \leq \rho\}$.

Lemma 4.2 (One regular compact window). *For every positive-area compact $K \subset Q$ and $0 < \varepsilon < \text{area}(K)$, there is a compact $T \subset K \setminus \{0\}$, with $\text{area}(T) > \text{area}(K) - \varepsilon$, and $L_T < \infty$ such that*

$$(4.5) \quad \text{area}(T \cap D_{w^-} \cap F_w^{-1}B) \leq L_T \text{area}(B)$$

for all $1 \leq |w| \leq J$ and all Lebesgue measurable $B \subset \mathbb{R}^2$. On the same set, for every $\eta \in (0, h)$ there is $D_\eta \geq 1$ such that every local exactly feasible tail v , after any exactly feasible J -prefix from T , satisfies

$$(4.6) \quad \begin{aligned} D_\eta^{-1} e^{-(h+\eta)|v|} &\leq \lambda_v \leq D_\eta e^{-(h-\eta)|v|}, \\ D_\eta^{-1} e^{-(\chi_R+\eta)|v|} &\leq P_v \leq D_\eta e^{-(\chi_R-\eta)|v|}. \end{aligned}$$

The set and L_T precede η , every input, tolerance and horizon.

Proof. For $d > 0$, the total weight $\sum \lambda_v$ of length- m words with $|\#0/m - a_0| > d$ is at most $2e^{-2d^2m}$. Indeed, for a Bernoulli variable X of mean a_0 , the second derivative of $\log \mathbb{E}e^{t(X-a_0)}$ is a tilted variance, at most $1/4$. Its value and derivative at zero vanish, so it is at most $t^2/8$. Exponential Markov with $t = 4d$ and $t = -4d$ gives the bound. By (3.7), the unions of the corresponding complete local feasible regions have summable area. Borel–Cantelli for rational $d > 0$ gives a Borel null set $B_{\text{loc}} \subset Q_\rho$, including the origin, outside which all feasible length- m prefixes have zero frequency tending uniformly to a_0 . Overlaps do not invalidate this union bound. Controlled invariance supplies at least one feasible prefix at every length.

Lemma 4.1 shows that

$$B_J = \bigcup_{|\sigma|=J} (D_\sigma \cap F_\sigma^{-1}B_{\text{loc}})$$

is null. The finite union $Z_J = \bigcup_{1 \leq |w| \leq J} Z_w$ is a compact null set. On $Q \setminus B_J$, the maximum frequency deviation over all feasible transient choices and all their feasible length- m tails tends to zero. It is measurable: for each m there are finitely many words and their complete feasibility domains are closed. Egorov on K , followed by removal of $B_J \cup Z_J \cup \{0\}$ and compact inner approximation, gives T of the stated area with uniform convergence. Allocate less than $\varepsilon/2$ loss to Egorov and less than $\varepsilon/2$ to inner approximation. Both logarithmic products are affine functions of frequency. Uniform convergence gives (4.6); finitely many early lengths are absorbed in D_η . This selects one set before all precision choices, rather than a set for each exponent.

For a word w with $1 \leq |w| \leq J$, the compact set $M_w = T \cap D_{w^-}$ is disjoint from Z_w . The composition F_w is therefore regular at every point of M_w . Cover it by finitely many open inverse charts $U_{w,k}$, with compact closures inside regular inverse charts. Change of variables on each chart gives

$$\text{area}(M_w \cap U_{w,k} \cap F_w^{-1}B) \leq \left(\sup_{\bar{U}_{w,k}} |\det DF_w|^{-1} \right) \text{area}(B).$$

The supremum is finite. Sum over the charts and take the maximum over the finite word family, omitting empty M_w 's, to obtain (4.5). Chart overlaps only enlarge the bound; every local inverse branch meeting M_w is included. $\square$

The inverse-area bound is confined to the selected regular window; its constant may depend on the prescribed area loss. In particular, it supplies no uniform inverse-Jacobian bound on all Q . The upper catalogues below still cover every actual state, including the critical states omitted from this window.

Proof of Theorem 2.5. We divide the covered states into three classes: those that exit during the transient, those that remain exactly feasible through the next m steps, and those that first exit during those m steps. These classes give, in order, the three terms of

(2.21). Choose J, T, L_T from Lemma 4.2. Let $Q_\delta = \{x : \text{dist}(x, Q) < \delta\}$. For $0 < \delta \leq 1$, the tubes around the three edges of Q give

$$\text{area}(Q_\delta \setminus Q) \leq C_Q \delta, \quad C_Q = 2(2 + 2\sqrt{2}) + 3\pi.$$

If a state in $T \cap A_u(n, \delta; K)$ first truly exits at $j \leq J$, put $w = u|j$. Its earlier states lie exactly in Q , and it belongs to

$$T \cap D_{w^-} \cap F_w^{-1}(Q_\delta \setminus Q).$$

By (4.5), its class has area at most $L_T C_Q \delta$. Summing over the J possible first-exit times gives

$$(4.7) \quad \text{area}\{x \in T \cap A_u(n, \delta; K) : \text{first true exit occurs by } J\} \leq C_T \delta, \quad C_T = J L_T C_Q.$$

All derivative inputs here are in Q ; no nonsingularity condition is needed at the last output, which may be outside.

For the remaining states, $\sigma = u|J$ is exactly feasible. Transport them by F_σ to Q_ρ , and use (4.5) to pull back endpoint area estimates. If the next m prescribed digits are exactly feasible for at least one transported state, (3.7) and (4.6) bound the area of the survival class by $C_{T,\eta} e^{-(h-\eta)m}$. If no such state exists, that class is empty. For a first true local exit at time $j \leq m$, the prescribed $(j-1)$ -prefix is feasible for a transported state whenever the class is nonempty. Its pulled-back area, by (3.9), (4.5) and both products in (4.6), is at most

$$C_{T,\eta} \delta e^{(\chi_R - h + 2\eta)(j-1)}.$$

Since $\chi_R > h$, the sum is bounded by a fixed multiple of $\delta e^{(\chi_R - h + 2\eta)m}$. Together with (4.7) and exact survival, this proves (2.21). A switch between feasible codes was never counted as an exit. Every covered state satisfies the constraint at its first true exit, regardless of subsequent reentry. $\square$

5. PHYSICAL TRAJECTORIES WITH A WRONG-CONTROL MARGIN

The area estimate concerns positive-area mass. For the full constraint we also need actual states whose controls cannot be replaced by approximate alternatives. We construct them at one fixed positive radius for each run bound, then force prefixes while their physical margin remains larger than the tolerance.

Lemma 5.1 (Bounded-run realisation). *For each integer $L \geq 2$, there is a positive radius s_L and $d'_L > 0$ such that, for every $0 < s \leq s_L$, every infinite binary sequence whose successive runs have length at most L is realised by an actual exactly feasible orbit starting at (s, sy_0) . At every step of that orbit the alternative control is genuinely infeasible, with an angular excess at least d'_L . The initial-angle centre can be chosen Lipschitz in s , with a constant uniform over the sequences. The radius bound and margin are uniform over the sequences. No such assertion is made for unbounded runs or all constant words.*

Proof. We first keep the reference angular orbit away from the boundary. We then solve the perturbed angular and radial recursions together near that orbit. This will preserve both feasibility and the margin against the alternative control. The inverse reference maps are $f_0(y) = a_0 y - a_1$, $f_1(y) = a_1 y + a_0$. For any fixed infinite input v , successive inverse compositions give a unique reference bounded orbit $y_k^0 \in [-1, 1]$. If runs have length at most L , the next opposite digit is seen within $L+1$ digits. Since $f_0(y) + 1 = a_0(y+1)$, $1 - f_1(y) = a_1(1-y)$, and each opposite digit inserts an endpoint distance at least $2a_*$,

$$(5.1) \quad |y_k^0| \leq 1 - d_L, \quad d_L = 2a_*^{L+1} > 0.$$

Let $b = d_L/4$ and consider the complete closed ball of sequences $\|y - y^0\|_\infty \leq b$ in ℓ^∞ . For each such sequence and a positive fixed initial radius s_* , recursively define $s_{k+1} = s_k R_{v_{k+1}}(s_k, y_k)$. These radii stay below $\bar{r}^k s_*$, independently of whether the angular

recursion has yet been imposed. The scalar map $s \mapsto sR_i(s, y)$ has derivative bounded in absolute value by $\hat{r} < 1$, and its y derivative is bounded by Bs^2 , since $R_i \leq \bar{r} < 1$ and (3.3) bounds its logarithmic angular derivative. Thus two candidate sequences give

$$(5.2) \quad \sup_k |s_k(y) - s_k(\tilde{y})| \leq C_s s_*^2 \|y - \tilde{y}\|_\infty, \quad C_s = B/(1 - \hat{r}).$$

Write $e_i(s, y) = T_i(s, y) - L_i(y)$. By (3.3), $|e_i| \leq Bs$, $|\partial_s e_i| \leq B$, and $|\partial_y e_i| \leq Bs$. Define

$$(5.3) \quad (\mathcal{B}y)_k = f_{v_{k+1}}(y_{k+1}) - a_{v_{k+1}} e_{v_{k+1}}(s_k(y), y_k).$$

Choose $s_* \leq \rho$ so small that

$$(5.4) \quad a^* B s_* \leq (1 - a^*)b/2, \quad a^* B(s_* + C_s s_*^2) < (1 - a^*)/2.$$

Then (5.3) maps the ball into itself and has Lipschitz constant at most $a^* + a^* B(s_* + C_s s_*^2) < 1$. The contraction mapping theorem gives a fixed sequence. Its fixed-point equation is $y_{k+1} = T_{v_{k+1}}(s_k, y_k)$, the required angular recursion. It has $|y_k| \leq 1 - 3d_L/4$, so the constructed physical trajectory is exactly feasible for all times.

To check the alternative control, suppose the chosen digit is zero. Then $T_0(s_k, y_k) = y_{k+1} \leq 1 - 3d_L/4$, and (3.3) gives

$$y_k \leq a_0 - a_1 - 3a_0 d_L/4 + a_0 B s_k.$$

Substitution into $T_1 = L_1 + e_1$ yields

$$T_1(s_k, y_k) \leq -1 - \frac{3a_0 d_L}{4a_1} + B(1 + a_0/a_1)s_k.$$

For chosen digit one, the symmetric computation gives

$$T_0(s_k, y_k) \geq 1 + \frac{3a_1 d_L}{4a_0} - B(1 + a_1/a_0)s_k.$$

Shrink s_* once more, before considering any particular sequence, so that these last error terms are at most $a_* d_L/(4a^*)$. Both wrong angular excesses are then at least $d'_L = a_* d_L/(2a^*)$. Set $s_L = s_*$. All choices depend only on L and the fixed system. The same estimates hold for every initial radius $0 < s \leq s_L$. For a fixed candidate sequence, scalar contraction gives $\sup_k |s_k(s, y) - s_k(s', y)| \leq |s - s'|$. If $\kappa_L < 1$ is the common contraction constant of (5.3), the fixed-point equations therefore give

$$(5.5) \quad \|y^v(s) - y^v(s')\|_\infty \leq \frac{a^* B}{1 - \kappa_L} |s - s'|.$$

□

The preceding margin is angular. Its physical size decreases with the radius, so a long exact word need not remain distinguishable at an assigned tolerance. The next lemma chooses a shorter prefix and compares its radial scale directly with that tolerance.

Lemma 5.2 (Forcing a block prefix). *Fix $L \geq 2$ and $1 \leq j \leq L - 1$, and put*

$$M_L = \binom{L-2}{j-1}, \quad c_L = -\frac{j}{L} \log r_0 - \left(1 - \frac{j}{L}\right) \log r_1.$$

For every precision sequence of exponent $\gamma > 0$ and every $0 < t < \min\{1, \gamma/c_L\}$, for all sufficiently large n ,

$$(5.6) \quad r_F(n, \delta_n; Q) \geq M_L^{\lfloor tn/L \rfloor}.$$

Proof. Use all length- L blocks starting with zero, ending with one, and containing j zeros. Every concatenation has runs of length at most L and radial product e^{-Lc_L} per complete block. Put $k_n = \lfloor tn/L \rfloor$ and $m_n = Lk_n$. For each of the $M_L^{k_n}$ initial concatenations, append one fixed allowable block forever. Lemma 5.1 realises these infinite inputs at the same positive radius s_L . Every prefix through time m_n has $P_{v|k} \geq P_{v|m_n} = e^{-c_L m_n}$.

If an arbitrary catalogue input first differs from a witness at $j' \leq m_n$, its preceding states agree with that witness. By (3.6), their radius at $j' - 1$ is at least $C_R^{-1} s_L e^{-c_L m_n}$. The wrong radial factor is at least $r_*/2$, and the alternative angular excess is at least d'_L . Its actual physical excess therefore obeys

$$(5.7) \quad |z_{j'}| - s_{j'} \geq E_L e^{-c_L m_n}, \quad E_L = \frac{s_L r_* d'_L}{2C_R} > 0.$$

The ratio to $2\delta_n$ has logarithmic rate $\gamma - c_L t > 0$. For large n , it violates (3.8), regardless of later reentry. Thus the approximate input must agree with all m_n witness digits. Distinct selected prefixes give distinct actual states: at their first different digit one input is feasible and the other has the proved wrong-control margin. One catalogue input covers at most one of them, proving (5.6). The block length, positive radius, and margin remain fixed throughout this horizon limit. Integer rounding and every $o(n)$ precision fluctuation are absorbed by the strict choice of t . $\square$

6. STOPPING CATALOGUES AND THE MATCHED SPECTRUM

We first construct catalogues by following feasible inputs until the radius is small enough for a common tail. This construction applies under all the standing assumptions. We then prove the matched spectrum using the physical prefix lower bound and the covered-area estimate. The reference tree provides an upper bound; its leaves need not all be realised by actual states.

Lemma 6.1 (A complete radial stopping catalogue). *Put $M = \max\{1, 4\sqrt{2}\Lambda C_R \rho\}$. For $N \geq 0$ and $0 < t \leq 1$, let $\mathcal{L}_P(N, t)$ be the full reference binary tree stopped at the first $P_w \leq t$, or at depth N , whichever occurs first. For $n \geq J$ and $0 < \delta \leq 1$,*

$$(6.1) \quad r_F(n, \delta; Q) \leq 2^J \# \mathcal{L}_P(n - J, \delta/M).$$

The catalogue can retain only transient/leaf pairs whose complete actual feasible initial region is nonempty.

Proof. For every actual initial state choose an exactly feasible infinite input. Its first J letters form a feasible transient ending in Q_ρ ; the subsequent letters reach a reference leaf. Keep every pair realised by an actual state, including singleton regions and boundary states. Before stopping the assigned states are exactly feasible. At a threshold leaf, (3.6) gives

$$s_{|w|} \leq C_R \rho P_w \leq C_R \rho \delta/M \leq \delta/(4\sqrt{2}\Lambda).$$

Since $V = \Lambda s$ on Q , appending the same infinite zero tail and using (2.5) keeps the Euclidean norm at most $\delta/4$ at every later time. The origin belongs to Q , so every constraint holds strictly on that tail. A horizon leaf is exactly feasible through time n , and its later extension is irrelevant. Every constructed input fixes the origin as well. There are at most 2^J transients, proving (6.1). $\square$

6.1. Upper bound for the matched spectrum. For the rest of this section assume $\beta_0 = \beta_1 = \beta$. Then $P_w = \lambda_w^\beta$, $\chi_R = \beta h$, and

$$t_\delta = (\delta/M)^{1/\beta}, \quad b_\delta = -\log t_\delta.$$

Lemma 6.1 uses the tree stopped at $\lambda_w \leq t_\delta$ or at depth $N = n - J$. Put $c_i = -\log a_i$, $c_{\min} = \min_i c_i > 0$, $c_{\max} = \max_i c_i$, and $\Gamma_{n,\delta} = \beta(b_\delta + c_{\max})/n$. A threshold leaf has logarithmic angular cost in $[b_\delta, b_\delta + c_{\max}]$; an unfinished horizon leaf has cost below b_δ . Thus every nonempty leaf of length $\ell \leq n$, containing j zeros with $p = j/\ell$, satisfies

$$(6.2) \quad \ell \chi(p) < b_\delta + c_{\max}, \quad \frac{\ell}{n} \leq \min\{1, \Gamma_{n,\delta}/[\beta \chi(p)]\}.$$

The corresponding term of the binomial law of parameter j/ℓ gives $\binom{\ell}{j} \leq e^{\ell H_b(j/\ell)}$. Endpoint types have one word. Each leaf type therefore has at most $e^{nS(\Gamma_{n,\delta})}$ words. There are at most $(n+1)^2$ length/type pairs, including a possible empty word. By the actual catalogue construction,

$$(6.3) \quad r_F(n, \delta; Q) \leq 2^J (n+1)^2 e^{nS(\Gamma_{n,\delta})}.$$

Only pairs realised by actual initial states are needed for this upper bound.

Every function under the maximum defining S is Lipschitz in γ , with common constant at most $\log 2/(\beta c_{\min})$. Their maximum is continuous. Here $\beta b_{\delta_n} = -\log \delta_n + \log M$, so $\Gamma_{n,\delta_n} \rightarrow \gamma$ for every precision sequence. The fixed transient and polynomial factors have zero rate, giving

$$(6.4) \quad \limsup_n n^{-1} \log r_F(n, \delta_n; Q) \leq S(\gamma).$$

This includes $\gamma = 0$ and nonmonotone sequences.

The reference angular leaf weights sum to one, and each is at least $a_* t_\delta$: a threshold leaf has parent weight above t_δ , whereas an unfinished leaf has weight above t_δ itself. This identity concerns a reference tree, not a physical partition. Enlarging a constant for $n < J$ yields the useful coarse bound

$$(6.5) \quad r_F(n, \delta; Q) \leq C_U \delta^{-1/\beta} \quad (0 < \delta \leq 1), \quad r_F(n, \delta; Q) \leq 2^n.$$

6.2. Shorter physical prefixes give the lower bound. Let $\gamma > 0$ and fix $p \in (0, 1)$. Choose $j_L \in \{1, \dots, L-1\}$ with $j_L/L \rightarrow p$, and put

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L).$$

The exact identity and elementary type bounds

$$\binom{L-2}{j-1} = \frac{j(L-j)}{L(L-1)} \binom{L}{j}, \quad \frac{e^{LH_b(j/L)}}{L+1} \leq \binom{L}{j} \leq e^{LH_b(j/L)}$$

give $B_L \rightarrow H_b(p)$ and $\chi_L \rightarrow \chi(p)$. For the lower type bound the j -th term is a largest term of the binomial law with parameter j/L , so is at least $1/(L+1)$. The additional fraction has logarithm $o(L)$ for interior p .

The radial block cost in Lemma 5.2 is now $c_L = \beta \chi_L$. Fix L first and take

$$(6.6) \quad 0 < t < \min\{1, \gamma/(\beta \chi_L)\}.$$

The lemma, which checks the physical margin and every constrained time, gives

$$(6.7) \quad \liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq t B_L.$$

For fixed L take the supremum over the strict t , next let $L \rightarrow \infty$, and finally take the supremum over $p \in (0, 1)$. Endpoint frequencies have zero entropy, so continuity yields $\liminf n^{-1} \log r_F(n, \delta_n; Q) \geq S(\gamma)$. No witness radius or margin is required to remain uniformly positive as $L \rightarrow \infty$; they are fixed in each horizon limit. For $\gamma = 0$, the bound $r_F \geq 1$ and (6.4) give the zero limit. This proves (2.15).

6.3. Positive-area and fixed near-full initial sets. To complete Theorem 2.3, apply Theorem 2.5 to a fixed positive-area compact K and its positive-area set T . A catalogue covering K has at least $\text{area}(T)/\sup_u \text{area}(T \cap A_u(n, \delta_n; K))$ elements. For fixed $\eta \in (0, h)$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + \eta} \right\rfloor \right\}.$$

Eventually $m_n \geq 0$, and $\delta_n \leq e^{-(\chi_R + \eta)m_n}$. Each term on the right of (2.21) is then at most a constant times $e^{-(h - \eta)m_n}$. Consequently

$$\liminf_n n^{-1} \log r_F(n, \delta_n; K) \geq (h - \eta) \min\{1, \gamma/(\chi_R + \eta)\}.$$

Since $\chi_R = \beta h$ here, letting $\eta \downarrow 0$ gives the lower bound in (2.16); for $\gamma = 0$ the bound $r_F \geq 1$ also suffices. The upper bound follows from $r_F(n, \delta; K) \leq r_F(n, \delta; Q)$ and (2.15). The evaluation of S below will give the claimed common low-side rate.

For $\zeta \in (0, 1)$ take the single near-full set T constructed in Section 4 and put $K_\zeta = T \cup \{0\}$. All exactly feasible length- n words meeting T have, after their J -digit transient, frequency uniformly tending to a_0 . If $d_N \rightarrow 0$ bounds this deviation for tails of length $N = n - J$, there are at most

$$2^J(N + 1) \exp\left(N \max_{|p - a_0| \leq d_N} H_b(p)\right) = e^{(h + o(1))n}$$

such words. Every state of T has an exactly feasible prefix, so these words, extended by any infinite tail, cover T through time n . One additional input covers the origin. This gives an upper rate h for every precision sequence, without selecting a strategy. Bound (6.5) also gives upper rate γ/β . The lower bound already proved gives (2.17). The construction of T used all-prefix frequency convergence, removal of null critical pullbacks, and compact inner approximation; it preceded every exponent, precision sequence and horizon. This completes Theorem 2.3 subject to the following elementary evaluation.

6.4. Evaluation of the maximum.

Proof of Corollary 2.4. Reparameterise the maximum by the frequency q of the smaller branch $a = a_*$. In this variable $\chi = c_b + d_b q$. Write $x = \gamma/\beta$. The entropy inequality

$$c_b + d_b q - H_b(q) = q \log(q/a) + (1 - q) \log((1 - q)/(1 - a)) \geq 0$$

follows from $-\log v \geq 1 - v$ and extends to the endpoints by continuity; equality holds at $q = a$. For $x \leq h$, the objective $H_b(q) \min\{1, x/(c_b + d_b q)\}$ is at most x and equals x at a .

For $h < x < c_b + d_b/2$ put $q_x = (x - c_b)/d_b \in (a, 1/2)$. When $q \leq q_x$ the objective is $H_b(q) \leq H_b(q_x)$. When $q \geq q_x$ it is $xH_b(q)/(c_b + d_b q)$. This ratio decreases for $q > a$: the numerator of its derivative is

$$D(q) = (c_b + d_b q)H'_b(q) - d_b H_b(q), \quad D(a) = 0, \quad D'(q) = (c_b + d_b q)H''_b(q) < 0.$$

The maximum is therefore $H_b(q_x)$. Finally, for $x \geq c_b + d_b/2$, the value at $q = 1/2$ is $\log 2$, the universal entropy maximum. Since $a < 1/2$, $c_b + d_b/2 > c_b + d_b a = h$. The three pieces match at the thresholds, proving (2.18)–(2.19) and the low-side assertion of Theorem 2.3. Subtracting (2.17) gives (2.20). Strict increase of H_b on $(a, 1/2)$ proves the stated strict gap and the exact saturation threshold. $\square$

7. PHYSICAL PROBABILITY AND EXPONENTIAL RELIABILITY

Throughout this section the powers match, $r_i = a_i^\beta$, and (2.11) holds. The proof has two parts. A uniform bound on probability transport lets us discard rare feasible words and obtain the L_a upper bound; the full-state construction already gives the S upper bound. For the reverse inequality we thicken the trajectories from Section 5 into disjoint regions of known mass. Their total mass determines which word families a successful catalogue must cover.

7.1. All feasible transient branches. Let J and ρ be as in Section 4, so every exactly feasible J -prefix ends in Q_ρ . Recall D_w , the complete exactly feasible domain, and w^- , the word with its last digit removed.

Lemma 7.1 (Whole-triangle probability domination). *There is a fixed $K_J < \infty$ such that for every $1 \leq |w| \leq J$ and measurable $B_0 \subset \mathbb{R}^2$,*

$$(7.1) \quad \mu(D_{w^-} \cap F_w^{-1} B_0) \leq K_J \text{area}(B_0).$$

Consequently, for every family $\mathcal{V}$ of words of a common length N ,

$$(7.2) \quad \mu\left(\bigcup_{\substack{|\sigma|=J \\ v \in \mathcal{V}}} D_{\sigma v}\right) \leq C_0 \sum_{v \in \mathcal{V}} \lambda_v, \quad C_0 = \max\{1, 2^J K_J C_A \rho^2\}.$$

The constants are independent of N , the word family, tolerance, failure budget, and horizon.

Proof. Compactness gives $d_0 = \min_{i,x \in Q} |\det DF_i(x)| > 0$. On D_{w^-} all inputs to the derivative factors of F_w belong to Q , even when its last output leaves Q . Thus $|\det DF_w| \geq d_0^{|w|}$ there. Cover this compact domain by finitely many local inverse charts for F_w , with compact closures inside regular charts. Change of variables bounds the pulled-back area in each chart by its inverse-determinant supremum times $\text{area}(B_0)$. Sum over the charts and maximise over the finitely many words of length at most J . All inverse branches meeting the domain are included; their overlap only enlarges the upper bound. No global inverse is used.

An exactly feasible σ sends D_σ into Q_ρ . If σv is feasible, the state after σ lies in the complete initial region for the tail v . Precisely,

$$D_{\sigma v} \subset D_\sigma \cap F_\sigma^{-1}(\mathcal{I}_v \cup \{0\}).$$

By (3.7), this target has area at most $C_A \rho^2 \lambda_v$. Its possible point at zero has null pullback by (7.1). Apply that bound and sum over all at most 2^J transients and all v to obtain (7.2). $\square$

The word regions may overlap; (7.2) only requires an upper bound on the probability of their union. This is enough for the upper construction below, which uses exactly feasible prefixes. In contrast, applying (7.1) to a transient first exit would leave an area error of $JK_J C_Q \delta$. Such an error cannot be charged to the failure budget, since δ_n may be exponentially larger than e_n .

7.2. Two catalogue upper bounds. To obtain the reliability bound, keep words whose total discarded probability is below e_n . This catalogue is exactly feasible through time n on every state it serves, so its construction has no tolerance error to subtract from the failure budget. Fix $\varepsilon > 0$, $n > J$, and $N = n - J$. Keep every exactly feasible length- n prefix σv , with $|\sigma| = J$, whose tail zero frequency p satisfies $I_a(p) \leq \alpha + \varepsilon$. Empty complete regions may be omitted. Append any infinite input to each kept horizon prefix. Every state not covered has an exactly feasible length- n prefix by controlled invariance, and all such prefixes from that state have discarded tails. By Lemma 7.1, its total probability is at most

$$(7.3) \quad C_0 \sum_{I_a(j/N) > \alpha + \varepsilon} \binom{N}{j} e^{-N\chi(j/N)} \leq C_0(N+1)e^{-N(\alpha+\varepsilon)} < e_n$$

for large n . Here $\binom{N}{j} \leq e^{NH_b(j/N)}$ follows from the j -th term of the binomial probability sum. The catalogue has at most

$$(7.4) \quad 2^J(N+1)e^{NL_a(\alpha+\varepsilon)}$$

members. The fixed transient and polynomial factors have zero rate. Continuity of L_a , proved below, gives upper rate $L_a(\alpha)$.

Independently, the full-state stopping construction in Lemma 6.1 gives upper rate $S(\gamma)$. It assigns an actual feasible prefix to each state; after a threshold leaf its common tail satisfies every remaining constraint. Taking the smaller of these two catalogue bounds yields

$$(7.5) \quad \limsup_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \leq \min\{S(\gamma), L_a(\alpha)\}.$$

7.3. A fixed positive mass gives the low-side lower bound. Take one regular compact $T \subset Q$ of area greater than $3/4$ from Theorem 2.5. For fixed $0 < \varepsilon < h$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\beta h + \varepsilon} \right\rfloor \right\}.$$

Then $\delta_n \leq e^{-(\beta h + \varepsilon)m_n}$ and, using $\chi_R = \beta h$, (2.21) bounds the area covered inside T by one input by a fixed multiple of $e^{-(h - \varepsilon)m_n}$. A successful catalogue covers more than $1/2$ area inside T eventually. The union bound and $\varepsilon \downarrow 0$ give

$$(7.6) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \min\{h, \gamma/\beta\}.$$

Since $S(\gamma) = \gamma/\beta$ for $\gamma \leq \beta h$ and $L_a(\alpha) \geq h$, this matches (7.5) on the entire low side. At $\gamma = 0$, the upper bound and $r_{e_n} \geq 1$ give the zero limit. Fixed loss is used here to retain a positive successful mass, not to construct an upper bound with exponentially small failure.

7.4. Positive-area regions forcing complete prefixes. The witnesses in Lemma 5.1 distinguish control words, but individual states have zero area. For a reliability lower bound we need a whole region for each word, with controlled mass and the same forced prefix at every point. The next lemma obtains these regions by thickening the witnesses on the angular scale λ_w .

Lemma 7.2 (Thickened physical words). *For every integer $L \geq 2$ there are $\tau_L, E_L > 0$ such that any family $\mathcal{W}$ of distinct words of a common length m , each admitting an infinite extension with runs at most L , has pairwise disjoint measurable regions $E_w \subset Q$ with*

$$(7.7) \quad \mu(E_w) = \tau_L \lambda_w.$$

Every point of E_w is exactly feasible under w . If $n \geq m$ and $2\delta < E_L P_w$, every input whose actual successful set $A_u(n, \delta; Q)$ meets E_w has prefix w . All constants are fixed before m, n, δ and the word family.

Proof. We choose the thickness so that the selected word remains feasible and each possible first wrong control retains an angular margin. After computing the area and proving disjointness, the radial product converts that margin into the required physical distance. For each word choose a bounded-run extension $v(w)$. Lemma 5.1 gives a common radius bound $s_L > 0$, continuous centres $y_w(s)$ for $0 < s \leq s_L$, and wrong-control excess $d'_L > 0$. Its construction gives centre angles at distance at least $\Delta_L = 3d_L/4$ from both boundaries at every step, where $d_L = 2a_*^{L+1}$. Put $K_0 = 1/a_* + 3B\rho$ and choose

$$0 < \epsilon_L \leq \min\{1/(2C_A), \Delta_L/(4C_A), d'_L/(4K_0C_A)\}.$$

Define

$$(7.8) \quad E_w = \{(s, sy) : s_L/2 \leq s \leq s_L, |y - y_w(s)| \leq \epsilon_L \lambda_w\}.$$

Fix the initial radius. On the feasible component between the centre and any feasible angle, (3.6) bounds the change in the angle after each prefix $w|j$ by

$$(7.9) \quad |Y_{w|j}(y) - Y_{w|j}(y_w(s))| \leq C_A \epsilon_L \lambda_w / \lambda_{w|j} \leq D_L, \quad D_L = C_A \epsilon_L.$$

If the complete interval $I_w(s)$ ended within the proposed initial interval, continuity would make an initial or intermediate angular constraint attain an endpoint. Its centre is at distance Δ_L , while $D_L \leq \Delta_L/4$, a contradiction. The radii stay positive and below $\rho < 1$, so no radial constraint terminates the interval. Thus the entire interval is feasible. This argument applies even when the complete terminal image is truncated; it does not assume that image is onto.

At each prefix the thickened state and its centre lie on one image graph with logarithmic radial slope at most one. Hence their log radius ratio has absolute value at most $D_L \leq 1/2$, and their radius difference is at most $2\rho D_L$. The alternative angular map changes by at most

$$(1/a_* + B\rho)D_L + B(2\rho D_L) = K_0 D_L \leq d'_L/4.$$

Every thickened point therefore retains a wrong-control excess at least $d'_L/2$ at every prefix, as well as selected-control feasibility. Continuity of the centres makes E_w measurable. Its actual area is

$$2\epsilon_L \lambda_w \int_{s_L/2}^{s_L} s ds = \frac{3\epsilon_L s_L^2}{4} \lambda_w; \quad \tau_L = 3\epsilon_L s_L^2/4.$$

Two distinct equal-length words cannot share a point in their regions: at their first difference one selected control would be wrong, contradicting feasibility under both words.

If an arbitrary input first differs at $j \leq m$, the preceding trajectory is exactly the selected one. Its radius is at least $C_R^{-1}(s_L/2)P_{w|j-1} \geq C_R^{-1}(s_L/2)P_w$. The wrong output has radial factor at least $r_*/2$ and angular excess at least $d'_L/2$. Thus

$$(7.10) \quad |z_j| - s_j \geq E_L P_w, \quad E_L = \frac{s_L r_* d'_L}{8C_R} > 0.$$

Either $z - s$ or $-z - s$ has this positive excess. Each function has Euclidean Lipschitz constant $\sqrt{2}$ and is nonpositive on Q , so the distance is at least $E_L P_w / \sqrt{2} > \delta$. The constraint is violated at that time. Later reentry cannot remove the violation. This proves the all-point forcing assertion. $\square$

7.5. Rare physical types and completion of the proof. Suppose $c = \gamma/\beta > h$. Fix an interior frequency p with

$$(7.11) \quad \chi(p) < c, \quad I_a(p) < \alpha.$$

The first strict inequality will make each selected prefix unavoidable at precision δ_n . The second will make the union of its physical regions too large to discard within e_n . We preserve both gaps while approximating the frequency by bounded-run blocks. Choose $j_L/L \rightarrow p$ and use the length- L blocks starting with zero, ending with one, and containing j_L zeros. Write

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L), \quad I_L = \chi_L - B_L.$$

The binomial bounds proved in Section 6 give $B_L \rightarrow H_b(p)$, $\chi_L \rightarrow \chi(p)$ and $I_L \rightarrow I_a(p)$. Fix L large enough that $\beta\chi_L < \gamma$ and $I_L < \alpha$. Put $k = \lfloor n/L \rfloor$ and $m = Lk$. The M_L^k concatenations, extended by a fixed allowed block, have runs at most L . Lemma 7.2 gives disjoint regions of equal mass and equal radial product,

$$\mu(E_w) = \tau_L e^{-\chi_L m}, \quad P_w = e^{-\beta\chi_L m}.$$

The physical margin exceeds $2\delta_n$ eventually because its logarithmic ratio has rate $\gamma - \beta\chi_L > 0$. One input can then cover points in at most one region. Hence R inputs cover at most $R\tau_L e^{-\chi_L m}$ of the union, whose total area is $M_L^k \tau_L e^{-\chi_L m}$. At most e_n of this area may be missed, so every successful catalogue satisfies

$$(7.12) \quad R \geq M_L^k - \frac{e_n}{\tau_L e^{-\chi_L m}} = M_L^k \left(1 - \frac{e_n}{\tau_L e^{-I_L m}} \right).$$

The parenthesis tends to one, since $\alpha > I_L$. Take the horizon limit with L fixed, then let $L \rightarrow \infty$. This proves

$$(7.13) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \sup_{\chi(p) < c, I_a(p) < \alpha} H_b(p) \quad (c > h).$$

To evaluate this lower bound, let $a = a_* < 1/2$ and use q for the frequency of the smaller branch. Define the relabelled functions

$$\hat{\chi}(q) = c_b + dq, \quad \hat{I}(q) = q \log(q/a) + (1-q) \log((1-q)/(1-a)),$$

where $c_b = -\log(1-a)$ and $d = \log((1-a)/a)$, with the endpoint convention $0 \log 0 = 0$. If $a_0 = a$, then $q = p$; otherwise $q = 1-p$. Thus $\hat{\chi}(q) = \chi(p)$, $\hat{I}(q) = I_a(p)$, and $H_b(q) = H_b(p)$. This relabelling changes neither the dynamics nor either variational maximum. Put $J_a = \hat{I}(1/2) > 0$. Then $h = \hat{\chi}(a) = H_b(a)$, and Corollary 2.4 gives the three pieces of S . To evaluate L_a , frequencies $q < a$ have entropy at most h and cannot improve on the feasible frequency a . If $q > 1/2$, reflection preserves entropy and lowers relative entropy:

$$H_b(1-q) = H_b(q), \quad \hat{I}(1-q) - \hat{I}(q) = d(1-2q) < 0.$$

It therefore suffices to optimise over $[a, 1/2]$. On this interval H_b increases, and $\hat{I}$ increases strictly away from its minimum at a . Consequently

$$(7.14) \quad L_a(\alpha) = \begin{cases} H_b(q_\alpha), & 0 < \alpha < J_a, \quad \hat{I}(q_\alpha) = \alpha, \quad a < q_\alpha < 1/2, \\ \log 2, & \alpha \geq J_a. \end{cases}$$

The inverse of this strictly increasing continuous restriction of $\hat{I}$ is continuous; at J_a , q_α tends to $1/2$. Thus L_a is continuous on $(0, \infty)$, including its saturation point. For $c > h$ set $q_c = \min\{1/2, (c - c_b)/d\}$, and set $q_\alpha = 1/2$ if $\alpha \geq J_a$. Every $a < q < \min\{q_c, q_\alpha\}$ satisfies (7.11). Approaching its endpoint in (7.13) yields

$$\liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq H_b(\min\{q_c, q_\alpha\}) = \min\{S(\gamma), L_a(\alpha)\}.$$

Together with (7.5) and (7.6), this proves Theorem 2.1. At the common-cost endpoint $c = h$, the proof uses the fixed-mass argument, while the saturation and reliability endpoints use strict types approaching from below. Integer rounding changes m by less than fixed L ; the transient changes N by fixed J . The strict gaps in (7.3) and (7.12) absorb every $o(n)$ fluctuation in both sequences. No constants are required to be uniform as $L \rightarrow \infty$, or over different fixed systems.

Corollary 7.3 (Linear and fixed-failure specialisations). *The matched linear pair has rate (2.12) for every stated pair of sequences. Under the weaker standing null-critical hypotheses with matched powers, each fixed $e \in (0, 1)$ instead satisfies*

$$(7.15) \quad \lim_n n^{-1} \log r_e(n, \delta_n; Q) = \min\{h, \gamma/\beta\}.$$

Proof. The linear pair is nonsingular on Q , so Theorem 2.1 applies. For fixed failure, choose the single compact set in Theorem 2.3 with area greater than $1 - e$; its catalogue gives the upper rate in (7.15). For the lower rate choose a regular window T of area greater than $(1 + e)/2$. A catalogue with failure at most e covers at least $\text{area}(T) - e > 0$ area there. The fixed-mass proof above, which needs only Theorem 2.5, gives the lower rate. $\square$

8. A TRANSIENT PRESERVING FULL COSTS BUT CHANGING RELIABILITY

We prove Theorem 2.2 by changing only the first step from Q . An angular homeomorphism sends relatively large source intervals to much smaller intervals whose states force different control blocks. The dynamics become linear immediately after this bijective transient, preserving the minimum number of controls for full coverage. The transient changes

the probability of the forced regions, however, and these probabilities will decay too slowly for the prescribed failure budget. The construction below verifies both its C^2 regularity and its null critical set.

8.1. A flat increasing Cantor interpolation. Fix the parameters of the theorem and put $\lambda = a_0 a_1$. The inverse reference maps are $f_0(y) = a_0 y - a_1$ and $f_1(y) = a_1 y + a_0$. For the control blocks 01 and 10, write

$$B_0 = f_0 \circ f_1 = \lambda y + a_0^2 - a_1, \quad B_1 = f_1 \circ f_0 = \lambda y + a_0 - a_1^2.$$

Their invariant interval has endpoints and length

$$(8.1) \quad A = \frac{a_0^2 - a_1}{1 - \lambda}, \quad B = \frac{a_0 - a_1^2}{1 - \lambda}, \quad \ell = B - A = \frac{2\lambda}{1 - \lambda}.$$

Indeed $A + 1 = 2a_0^2/(1 - \lambda) > 0$ and $1 - B = 2a_1^2/(1 - \lambda) > 0$, so $[A, B] \subset (-1, 1)$. The maps take this interval to its left and right portions of relative length $\lambda < 1/2$. The corresponding Cantor set $\mathcal{C}$ consists of angles with infinite block controls 01, 10.

On $J_0 = [-1/2, 1/2]$ use the source maps

$$b = 2/5, \quad h_0(y) = by - (1 - b)/2, \quad h_1(y) = by + (1 - b)/2.$$

The two contraction ratios serve different purposes. The gap $\lambda < b^2$ will make the interpolation C^2 at Cantor points. At level k , the source intervals retain total length $(2b)^k$; this is the mass scale used in the reliability obstruction. Their Cantor set $\mathcal{D}$ is null, because the total level- k interval length is $(2b)^k \rightarrow 0$. For $\sigma \in \{0, 1\}^k$ put

$$J_\sigma = h_{\sigma_1} \circ \cdots \circ h_{\sigma_k}(J_0), \quad Z_\sigma = B_{\sigma_1} \circ \cdots \circ B_{\sigma_k}([A, B]).$$

Within each family these closed intervals are separated by positive gaps. Their lengths are b^k and $\ell\lambda^k$, respectively. Both address maps preserve order and are unique, including endpoints. The target Cantor set is null as $(2\lambda)^k \rightarrow 0$.

We now construct a strictly increasing C^2 homeomorphism $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$(8.2) \quad g(y) = y \quad (|y| \geq 1), \quad g([-1, 1]) = [-1, 1], \quad g(J_\sigma) = Z_\sigma, \quad g'(y) = 0 \iff y \in \mathcal{D}.$$

Let $E(t) = e^{-1/t}$ for $t > 0$ and zero otherwise, and set

$$\phi(t) = E(t)E(1 - t), \quad \Sigma(t) = \frac{\int_0^t \phi(v) dv}{\int_0^1 \phi(v) dv} \quad (0 \leq t \leq 1),$$

extended by zero and one outside that interval. The function is smooth, strictly increasing on $(0, 1)$, and flat at both endpoints. Map each point of $\mathcal{D}$ to the point of $\mathcal{C}$ with the same address. On a source gap (u, v) whose corresponding target gap is (U, V) , define

$$(8.3) \quad g(y) = U + (V - U)\Sigma((y - u)/(v - u)).$$

The ordered endpoint correspondence and increasing interpolation give the full interval identity $g(J_\sigma) = Z_\sigma$. At level k , the gap lengths are $(1 - 2b)b^k$ and $(1 - 2\lambda)\ell\lambda^k$. For $j = 1, 2$ the gap derivatives obey

$$(8.4) \quad |g^{(j)}| \leq C_j(\lambda/b^j)^k.$$

Since $\lambda < 2^{-10} < b^2$, both bounds tend to zero with level. Define q_1, q_2 by these derivatives inside gaps and by zero on $\mathcal{D}$. They are continuous, by (8.4) for high-level gaps and endpoint flatness for the remaining finite collection.

These candidate derivatives are actual derivatives. The integral of q_1 over a gap is its target gap length. Nullity of $\mathcal{C}$ and order preservation therefore imply $g(x) = A + \int_{-1/2}^x q_1(y) dy$ on J_0 ; inside a gap include the partial integral of (8.3). Moreover

$$\sum_{\text{gaps}} \int |q_2| \leq C \sum_{k \geq 0} (2\lambda/b)^k < \infty.$$

Each complete-gap integral of q_2 is zero, and a partial-gap integral equals $q_1(x)$. Since $\mathcal{D}$ is null, $q_1(x) = \int_{-1/2}^x q_2(y) dy$. The fundamental theorem of calculus gives $g' = q_1$ and $g'' = q_2$. Thus g is C^2 on J_0 , flat to second order on $\mathcal{D}$, with positive first derivative on all gaps.

For completeness, the exterior joins can be made without altering these properties. On $[L, R]$ with prescribed increment $T > 0$, set $d = \min\{(R - L)/4, T/4\}$ and $v(x) = E(x - L)E(R - x)$. For the left join use $c(x) = 1 - \Sigma((x - L)/d)$, and for the right use $c(x) = \Sigma((x - R + d)/d)$. In either case $\int c \leq d < T$. Integrate the positive interior density

$$(8.5) \quad q(x) = c(x) + \frac{T - \int_L^R c}{\int_L^R v} v(x).$$

For $(-1, -1/2, A + 1)$ start at $g(-1) = -1$; for $(1/2, 1, 1 - B)$ start at $g(1/2) = B$. The identity endpoint has derivatives 1, 0, and the Cantor endpoint has derivatives 0, 0. Extending by identity beyond $[-1, 1]$ therefore gives a global C^2 map satisfying (8.2). It is strictly increasing and onto, and $|g(y) - y| \leq 2$ globally. The map is fixed independently of every horizon and both sequences.

8.2. The maps and their dynamical properties. Set

$$\psi(s) = \frac{E(s - 1/4)}{E(s - 1/4) + E(1/2 - s)}.$$

Define

$$H(s, z) = \begin{cases} (s, z), & s \leq 1/4, \\ (s, z + s\psi(s)[g(z/s) - z/s]), & s > 1/4, \end{cases}$$

and set $F_i = A_i \circ H$. The smooth function ψ is zero on $s \leq 1/4$ and one on $s \geq 1/2$. The quotient is used only away from zero; its needed derivatives are locally bounded at every joining point, and flatness of ψ gives global C^2 maps. Near the origin H is identity, so the required first derivatives agree exactly.

On every fibre the angular map $g_s(y) = (1 - \psi(s))y + \psi(s)g(y)$ is an increasing homeomorphism of $[-1, 1]$ onto itself. Therefore $H(Q) = Q$, and one of the two linear controls is feasible at $H(x)$ for each $x \in Q$. Its radial factor is less than one. This proves controlled invariance including every boundary and the origin.

Put $m_* = \max_i \{a_i^\beta, a_i^{\beta-1}\} < 1$, $\Lambda = 1 + 10m_*/(1 - m_*)$, and $V = \max\{\Lambda|s|, |z|\}$. The bound $|H^z - z| \leq 2|s|$ gives, on the whole plane,

$$V(Hx) \leq (1 + 2/\Lambda)V(x), \quad V(A_i x) \leq m_*(1 + 1/\Lambda)V(x).$$

Consequently the common contraction factor can be taken as

$$\theta = m_*(1 + 1/\Lambda)(1 + 2/\Lambda) \leq m_*[1 + 5/\Lambda] < (1 + m_*)/2 < 1.$$

This applies to arbitrary input tails outside Q as well.

For $s > 1/4$ the determinant is

$$\det DH = 1 - \psi(s) + \psi(s)g'(z/s), \quad \det DF_i = a_i^{2\beta-1} \det DH;$$

for $s \leq 1/4$ the first determinant equals one. Thus the critical set in Q is exactly $\{(s, sy) : 1/2 \leq s \leq 1, y \in \mathcal{D}\}$. It has area zero by Fubini in physical coordinates. Local inverse charts off that null set show that every null target set has a null pullback in Q . No positive-area set is collapsed to a null set. Each fibre map is strictly increasing and identity outside $[-1, 1]$, so H is globally bijective. It is proper, since its first coordinate is unchanged and $|H^z - z| \leq 2|s|$. Thus it and both F_i are global homeomorphisms, although their Jacobian determinants vanish on the displayed null set.

The fixed radial factors satisfy

$$a_1^\beta = (1 - 2^{-10})^{4096} < e^{-4} < 1/4, \quad a_0^\beta < 1/4.$$

After the first input from Q , the radius lies below $1/4$, regardless of whether its angle is feasible. Every subsequent input therefore uses the identity part of H , even along an approximate trajectory that has left Q . For every input and every $j \geq 1$,

$$(8.6) \quad F_u^j x = A_u^j Hx \quad (x \in Q).$$

At time zero both x and Hx belong to Q . It follows that $A_u^F(n, \delta; Q) = H^{-1}A_u^A(n, \delta; Q)$ on Q . Since $H(Q) = Q$, full covers correspond, proving (2.13). Probability covers correspond only after transporting μ to $H_*\mu$, which is absolutely continuous but is not uniform area. Measure-transport invariance therefore compares the linear system with this transported measure, rather than with uniform area.

8.3. Positive-area forced families and the strict rate gap. Every infinite sequence of blocks 01, 10 has runs at most two. The reference endpoint-distance identities used in (5.1) give $|y_j| \leq 1 - d$ at all shifted positions, with $d = 2a_*^3$. For a chosen zero, the current angle is at most $a_0 - a_1 - a_0d$, so the alternative one gives at most $-1 - a_0d/a_1$; the symmetric bound holds for a chosen one. Thus every wrong-control angular excess is at least $d_* = (a_*/a^*)d > 0$. This holds on each complete interval Z_σ through its $2k$ -digit block prefix: its endpoints have such bounded-run extensions, and the linear prefix maps are monotone. No symbolic Cantor subset replaces the whole interval in this assertion.

For each $\sigma \in \{0, 1\}^k$ define an actual physical region

$$E_{\sigma,k} = \{(s, sy) : 1/2 \leq s \leq 1, y \in J_\sigma\}.$$

Here $\psi = 1$ and $g(J_\sigma) = Z_\sigma$. The regions are disjoint and each has mass $(3/8)b^k$, so their union has mass $(3/8)(2b)^k$. Every point requires its selected $2k$ -digit control prefix at sufficiently small tolerance. To check this for an arbitrary input, let its first differing digit be $j \leq 2k$. Equation (8.6) and $s \geq 1/2$ give the preceding physical radius at least $\frac{1}{2}\lambda^{\beta k}$: a partial block product before $2k$ is no smaller than the complete product λ^k . The wrong output radial factor is at least a_*^β and its angular excess is at least d_* . Therefore

$$(8.7) \quad |z_j| - s_j \geq c_0 \lambda^{\beta k}, \quad c_0 = \frac{1}{2} a_*^\beta d_* > 0.$$

Its distance from Q is at least this value divided by $\sqrt{2}$. For $k = \lfloor n/2 \rfloor$ and $\gamma = -\beta \log \lambda$, the logarithmic ratio to δ_n has positive rate $-\frac{1}{2}\beta \log \lambda$. Every successful input thus has the full selected prefix, for every point in the region. Later reentry cannot remove the constraint at this first wrong step.

One catalogue input can consequently cover points of at most one region. A catalogue of R inputs with failure at most e_n obeys

$$(8.8) \quad R \geq 2^k \left(1 - \frac{8e_n}{3(2b)^k} \right).$$

Since $-\frac{1}{2} \log(2b) = \frac{1}{2} \log(5/4) < 1/8 = \alpha$ by $\log(1+x) < x$, the parenthesis tends to one. Thus the lower limit is at least $\frac{1}{2} \log 2$. Both strict exponent gaps absorb all the allowed $o(n)$ fluctuations.

Finally $S(\gamma) = \log 2$: at $p = 1/2$ the threshold ratio $\gamma/[\beta\chi(p)]$ is two. For $q = 1/16$,

$$H_b(q) = \frac{1}{4} \log 2 + \frac{15}{16} \log(16/15) < \frac{1}{4} \log 2 + \frac{1}{16} < \frac{1}{2} \log 2,$$

and

$$I_a(q) > \frac{5}{8} \log 2 - H_b(q) > \frac{3}{8} \log 2 - \frac{1}{16} > \frac{1}{8},$$

using $\log 2 > 1/2$. Relative entropy is strictly increasing for $q > a_0$; its sublevel $I_a(q) \leq 1/8$ therefore has right endpoint below $1/16$. Binary entropy is increasing up to $1/2$, so $I_a(1/8) < H_b(1/16) < \frac{1}{2} \log 2$. This proves the strict inequality (2.14) and completes Theorem 2.2.

The example leaves the full-state finite-horizon counts unchanged but changes the probability weights of forced physical regions. Its exact reliability spectrum and the existence

of its reliability limit are not determined here; the strict lower-limit gap suffices for the obstruction. Absolute continuity and null-set pullback preservation alone do not control exponential tail masses.

9. A NONLINEAR EXAMPLE WITH INWARD BOUNDARY IMAGES

In this section the powers match: $\beta_0 = \beta_1 = \beta > 1$, so $r_i = a_i^\beta$ and $t_i = a_i^{\beta-1}$. The following example explains why the proof retained complete feasible intervals with possibly truncated images. Its branches overlap, yet long constant words have empty sections at a fixed positive radius. The maps are nonsingular and satisfy the reliability theorem. Their boundary images also rule out a constraint-preserving simultaneous conjugacy to the matched linear pair.

9.1. Empty fixed-radius slices. Let $t = \max a_i^{\beta-1} < 1$, $\Lambda = 1 + 4t/(1-t)$, $v = t(1 + 1/\Lambda) < 1$, and fix

$$0 < c < \min\{1 - a^*, (1 - v)/v, 1\}, \quad g(s) = 1 - \frac{cs}{1 + s^2}.$$

Consider the family

$$(9.1) \quad \begin{aligned} F_0(s, z) &= (r_0 s, g(s)t_0(z + a_1 s)), \\ F_1(s, z) &= (r_1 s, g(s)t_1(z - a_0 s)). \end{aligned}$$

Both are global smooth diffeomorphisms: $g \in [1 - c/2, 1 + c/2]$ is positive, the first coordinate has the explicit inverse $s = s'/r_i$, and the second is an affine invertible function of z at that radius. Their first jets are (2.4). For the stated V ,

$$(9.2) \quad V(F_i x) \leq \max\{r^*, (1 + c/2)v\} V(x), \quad \max\{r^*, (1 + c/2)v\} < 1.$$

This verifies the required common norm inequality on the whole plane.

For $0 < s \leq 1$, $a^* < g(s) < 1$. The complete one-step angular domains are

$$(9.3) \quad D_0(s) = [-1, a_0/g(s) - a_1], \quad D_1(s) = [a_0 - a_1/g(s), 1].$$

The two interior cut points lie strictly inside $(-1, 1)$, and the overlap has angular width $1/g(s) - 1 > 0$, physical width $s(1/g(s) - 1) = O(cs^2)$ near zero. Together with nonsingularity and (9.2), this verifies the assumptions of Theorems 2.3 and 2.1.

Yet for every fixed $0 < s_0 \leq 1$, a sufficiently long all-zero word is infeasible for every initial angle. To verify this, write $s_k = r_0^k s_0$ and $q_k = y_k + 1$. Under zero input,

$$(9.4) \quad q_{k+1} = \frac{g(s_k)}{a_0} q_k + 1 - g(s_k).$$

The multiplier is uniformly greater than one, since $g(s) > a^*$, and $q_1 \geq 1 - g(s_0) > 0$ for $y_0 \geq -1$. The sequence eventually exceeds two, violating $y \leq 1$. Monotonicity means that this also holds for the smallest initial angle, hence for all of them. The complete fixed-radius section is therefore empty for sufficiently long all-zero words.

This statement concerns $I_{0^m}(s_0)$ at a fixed radius, not the whole physical region $\mathcal{I}_{0^m}$. Choosing the radius smaller can make a given finite word feasible. Indeed any finite word admits a bounded-run infinite extension, and Lemma 5.1 realises that extension at a sufficiently small positive radius. Thus every finite word is realised somewhere in Q , despite the empty sections at larger radii.

There is also a constraint-preserving conjugacy obstruction. For (9.1), F_0 sends the positive lower edge of Q strictly inside Q , and its upper edge outside; F_1 does the symmetric thing. The image of the top edge has constant radius $r_i < 1$ and crosses the opposite angular boundary exactly once. Thus

$$(9.5) \quad F_i(\partial Q) \cap \partial Q \text{ consists of exactly two points.}$$

In contrast, for any target pair whose zero control sends the whole lower edge of Q into ∂Q and whose one control sends the whole upper edge into ∂Q , both corresponding intersections contain a continuum. A homeomorphism H of the plane with $H(Q) = Q$, simultaneously conjugating the two labelled controls (even allowing a label permutation), would map each intersection in (9.5) onto the corresponding target intersection, which is impossible because a homeomorphism preserves finiteness. Thus this example is not reducible to such a boundary-edge-preserving pair by a fixed Q -preserving simultaneous control homeomorphism, and in particular not by such a C^1 or bi-Lipschitz map. This obstruction concerns the stated example and conjugacies that preserve the constraint.

For example, take $a_0 = 1/3$, $a_1 = 2/3$, $\beta = 2$ and $c = 1/100$. The choice $\Lambda = 9$ in (9.2) gives a global factor at most $67/90 < 1$. Corollary 2.4 evaluates this same inward family at exponent $\gamma = \log 4$:

$$2 \left(\log 3 - \frac{2}{3} \log 2 \right) < \log 4 < \log(9/2), \quad \lim_n n^{-1} \log r_F(n, \delta_n; Q) = H_b \left(\frac{\log(4/3)}{\log 2} \right).$$

This holds for every precision sequence of that exponent. The first inequality follows from $(4/3)^3 > 2$ and the second from $4 < 9/2$. The spectrum therefore remains valid despite the empty fixed-radius sections.

10. CONSEQUENCES AND SCOPE

The distinction between ordinary precision cost and exponential reliability is quantitative. For the former, deleting an arbitrarily small fixed area leaves a compact set on which transient inverse area bounds suffice. For the latter, the discarded area would have to fit the shrinking failure budget. Nonsingularity on Q avoids this loss by giving uniform probability domination on the whole triangle. The Cantor transient shows why null-set preservation alone is insufficient, even when every full-state finite-horizon catalogue count is preserved.

The fixed-set and probabilistic conclusions concern different covering requirements. A single nearly full-area compact set has rate $\min\{h, \gamma/\beta\}$ at every finite exponent; a probability catalogue must cover mass $1 - e_n$ on the same Q , although its successful subset may vary with n . This distinction is essential above the common-cost threshold.

The scope remains the stated binary planar first-order family, controlled invariance of the triangle and common contraction toward the origin. The inward-boundary example allows actual overlap and truncated feasible images within this class. Nonsingularity is a sufficient reliability condition; the counterexample does not classify all maps with critical points. The high-precision mismatch spectrum, the high-side rate of an arbitrary compact initial set, and the exact reliability spectrum of the counterexample are not determined here.

APPENDIX A. MISMATCH AND THE COMMON-COST CRITERION

The main probability results use matched radial powers. This appendix asks whether matching is necessary for a common low-precision rate across all positive-area compact initial sets. The geometric estimates already retained the two products λ_w and P_w separately; only the stopping count changes when the powers differ. For this purpose define

$$\gamma_0 = -\log \max_i r_i > 0, \quad r_0^D + r_1^D = 1.$$

The root D is unique in $(0, 1)$, since the left side decreases from two to zero and $r_0 + r_1 < 1$. Also $\chi_R \geq \gamma_0$.

Theorem A.1 (First-order matching criterion). *For every fixed system satisfying the standing assumptions, the following property holds if and only if $\beta_0 = \beta_1$: there are $\gamma_1 > 0$*

and a function $c : (0, \gamma_1) \rightarrow [0, \infty)$ such that, for every fixed positive-area compact $K \subset Q$, every $\gamma \in (0, \gamma_1)$, and every precision sequence of exponent γ ,

$$\lim_n n^{-1} \log r_F(n, \delta_n; K) = c(\gamma).$$

Theorem A.2 (Low-precision mismatch). *Suppose $\beta_0 \neq \beta_1$. For each $\zeta \in (0, 1)$ there is one compact $K_\zeta \subset Q$, of area greater than $1 - \zeta$, such that simultaneously for every $0 < \gamma < \gamma_0$ and every precision sequence of that exponent,*

$$(A.1) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi_R < \gamma D.$$

The set is selected before the exponent, sequence, and horizon. Every other fixed positive-area compact $K \subset Q$ has lower rate at least $\gamma h / \chi_R$ in this range.

Proof of Theorem A.2. First consider the upper rate on Q . For every $0 < \gamma < \gamma_0$, all branches of the tree in Lemma 6.1 cross $P_w \leq \delta_n / M$ before depth $n - J$ for large n , since

$$P_w \leq e^{-\gamma_0(n-J)} < \delta_n / M \quad \text{at that depth.}$$

The ratio has logarithmic rate $\gamma - \gamma_0 < 0$, so the statement includes every $o(n)$ tolerance fluctuation. For the uncapped radial tree, write $N(t)$ for its leaf count. Replacing a parent by its children preserves $\sum P_w^D = 1$, and each leaf satisfies $r_* t < P_w \leq t$. Hence

$$(A.2) \quad t^{-D} \leq N(t) \leq (r_* t)^{-D}, \quad r_F(n, \delta_n; Q) \leq 2^J N(\delta_n / M).$$

This gives the upper rate γD with a catalogue satisfying every intermediate constraint. Optimality follows from the lower bound below.

For the lower rate put $p_* = r_0^D$, so $1 - p_* = r_1^D$, and $c_R(p) = -p \log r_0 - (1 - p) \log r_1$. Then

$$(A.3) \quad H_b(p_*) = D c_R(p_*).$$

Choose $j_L / L \rightarrow p_*$ and use the blocks in Lemma 5.2, with $c_L = c_R(j_L / L)$ and $B_L = L^{-1} \log \binom{L-2}{j_L-1}$. The binomial calculation in Section 6 gives $B_L \rightarrow H_b(p_*)$ and $c_L \rightarrow c_R(p_*)$. For each fixed L , $c_L \geq \gamma_0 > \gamma$. Every strict $0 < t < \gamma / c_L < 1$ therefore gives lower rate $t B_L$. Take its supremum first, then let $L \rightarrow \infty$. Equation (A.3) gives the lower rate γD . The positive-radius witnesses and wrong-control margins are those of the common physical lemma, not midpoints of assumed full images. This proves the first limit in (A.1).

For any fixed positive-area compact K , take the set T in Theorem 2.5. Fix $0 < \eta < h$ and set

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + \eta} \right\rfloor \right\}.$$

For large n it is nonnegative, and $\delta_n \leq e^{-(\chi_R + \eta)m_n}$. Every term of (2.21) is then bounded by a constant times $e^{-(h-\eta)m_n}$. Covering the positive area of T implies lower rate $\gamma(h - \eta) / (\chi_R + \eta)$, since $\gamma < \gamma_0 \leq \chi_R$. Let $\eta \downarrow 0$. This proves the asserted universal lower bound $\gamma h / \chi_R$.

For $K = Q$, choose the single set T of area greater than $1 - \zeta$ in the area proof, and put $K_\zeta = T \cup \{0\}$. Keep only stopped pairs in Lemma 6.1 realised on T . They cover it with the complete tails already constructed. Let $q_\delta = -\log(\delta / M)$ and $c_{\max} = -\log r_*$. Each retained tail of length ℓ has

$$q_\delta \leq -\log P_w < q_\delta + c_{\max}, \quad \ell \leq \frac{q_\delta + c_{\max} + \log D_\eta}{\chi_R - \eta},$$

by (4.6). At this length, $\lambda_w \geq D_\eta^{-1} e^{-(h+\eta)\ell}$, while all reference length- ℓ weights sum to one. There are at most $D_\eta e^{(h+\eta)\ell}$ such words. Sum over these lengths and the at most 2^J

transients, and include one input for the origin. For fixed $\eta \in (0, h)$ this gives

$$(A.4) \quad \log r_F(n, \delta_n; K_\zeta) \leq \frac{h + \eta}{\chi_R - \eta} [q_{\delta_n} + c_{\max} + \log D_\eta] + O_\eta(\log(1 + q_{\delta_n})) + O_\eta(1).$$

Because $q_{\delta_n} = \gamma n + o(n)$, letting $\eta \downarrow 0$ gives upper rate $\gamma h / \chi_R$, matching the lower bound. The set was selected through uniform frequency convergence before η , the precision exponent, the sequence, and the horizon.

Finally let $b_i = r_i^D$. The classical relative-entropy identity is

$$D\chi_R - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

Concavity of the logarithm proves the inequality, with equality only when $b_i = a_i$ for both labels. Equality would require $\beta_0 D = \beta_1 D = 1$, contrary to mismatch. Thus $D > h/\chi_R$, completing the theorem. $\square$

Proof of Theorem A.1. If $\beta_0 = \beta_1 = \beta$, Theorem 2.3 gives the common rate $c(\gamma) = \gamma/\beta$ on $(0, \beta h)$. If $\beta_0 \neq \beta_1$, Theorem A.2 separates Q and one fixed nearly full-area compact set for every $0 < \gamma < \gamma_0$. Any interval adjacent to zero contains such an exponent, contradicting the common-rate property. $\square$

The criterion concerns precisely the first derivatives (2.4), nonsingularity almost everywhere in Q , controlled feasibility, and the common norm inequality. The proof does not extend this necessity statement to arbitrary feedback systems, classify high-precision mismatch rates, or compare a fixed strategy with the optimal catalogue.

APPENDIX B. FINITE-HORIZON COUNTS AND THE CONTROL ALPHABET

This finite-horizon example separates equality of exponential rates from equality of catalogue counts when the available controls change. Let $q \geq 2$ be an integer, $0 < \rho < 1$, and consider

$$f_v(s, z) = (\rho s, \rho(qz + sv)), \quad U_q = \{q - 1 - 2j : 0 \leq j < q\}, \quad \widehat{U}_q = [-(q - 1), q - 1].$$

Use the same triangle Q , and put $K_1 = [1/2, 3/4] \times [-1/4, 1/4] \subset \text{int } Q$. Here $r_U^{\text{ex}}(n; K)$ denotes the minimum number of input words that keep every assigned state of K in Q exactly at times $0, \dots, n$, using alphabet U . This illustrative family is separate from the unequal-branch standing assumptions.

For either alphabet,

$$r_U^{\text{ex}}(n; Q) = q^n, \quad q^n/2 \leq r_U^{\text{ex}}(n; K_1) \leq q^n.$$

Indeed, at positive radius the angular map is $y \mapsto qy + v$. For the digit alphabet the inverse one-step intervals tile $[-1, 1]$, and their compositions produce q^n exactly feasible intervals of length $2q^{-n}$, including every intermediate constraint. They give the upper bounds for both alphabets. For any fixed word over the larger alphabet, the final angular map has slope q^n . Its feasible initial angles therefore have length at most $2q^{-n}$. Applying this bound to all of $[-1, 1]$, or to the slice $s = 1/2$ of K_1 with angles $[-1/2, 1/2]$, gives the two lower bounds. Consequently both exact exponential rates are $\log q$ for both alphabets.

The minimum count on K_1 can nevertheless decrease. For $q = 2$ and $n = 1$, input 1 fails on the positive-angle end of K_1 , and input -1 fails on its negative-angle end. Two digit controls are necessary and sufficient. Over $[-1, 1]$, input 0 sends every angle of K_1 to $2y \in [-1, 1]$ and keeps the actual radial coordinate in $(0, 1)$. Thus

$$r_{\{-1, 1\}}^{\text{ex}}(1; K_1) = 2, \quad r_{[-1, 1]}^{\text{ex}}(1; K_1) = 1.$$

Enlarging the control set preserves the displayed full-triangle counts and rate bounds, not all finite-horizon counts.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

Generative AI, including ChatGPT (OpenAI), was used extensively in the research and preparation of this draft. Its uses included developing mathematical arguments and examples, checking proof chains and source comparisons, and drafting, restructuring and compiling the English LaTeX manuscript. The research arguments are given in full in the text; AI-assisted checks do not constitute independent verification by a human author. Human authorship, affiliations, the extent of human mathematical review, and approval of the manuscript and this disclosure have not yet been confirmed. These facts must be completed by the responsible human authors before submission.

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